

CHARCHADESK

A Mini Project Report Submitted

**In
COMPUTER SCIENCE & ENGINEERING
DEPARTMENT**

By

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DECLARATION

We hereby declare that the work presented in this project report entitled “**CharchaDesk-Intelligent Real-Time Chat Application**” has been carried out by us as a part of our academic curriculum. This work is an original contribution and has been completed under the guidance and supervision of our project mentor. The matter embodied in this report has not been submitted, either wholly or in part, for the award of any other degree or diploma to any University or Institute.

We further declare that this project is the result of our own investigation, development, and implementation work. All sources of information, including research papers, articles, books, online resources, and technical documentation, have been properly acknowledged and cited wherever used. Any ideas, concepts, algorithms, diagrams, or code segments that are not originally developed by us have been appropriately referenced to their respective authors.

This project, “CharchaDesk – Intelligent Real-Time Chat Application”, involves the design and development of a real-time communication platform integrating modern web technologies such as Django, PostgreSQL, and WebSockets, along with intelligent features like AI-based chat summarization and smart reply suggestions. The implementation, testing, and results presented in this report are genuine and have been obtained through systematic experimentation and development.

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This report represents the original work carried out by the students during the academic session and has been found satisfactory as per the requirements of the institute.

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ABSTRACT

This report presents **CharchaDesk** is a **Real-Time Chat Application**, a modern communication platform designed to enhance user interaction by integrating real-time messaging with intelligent and productivity-oriented features. Traditional chat applications primarily focus on message exchange but often lack mechanisms to manage distractions and provide smart assistance to users.

The proposed system introduces a secure and scalable chat application developed using advanced technologies such as **Django for backend development, PostgreSQL for database management, and WebSockets for real-time communication**. The frontend is designed using modern web technologies to provide a responsive and user-friendly interface.

CharchaDesk incorporates several advanced features including **one-to-one real-time messaging, file sharing, message reactions, and user safety mechanisms such as blocking and reporting**. A key highlight of the system is the **Focus Mode**, which enables users to prioritize important conversations by reducing unnecessary notifications, thereby improving productivity.

In addition, the system integrates **Artificial Intelligence techniques**, including Natural Language Processing (NLP), to provide features such as **chat summarization and smart reply suggestions**. These intelligent functionalities help users save time, improve communication efficiency, and enhance overall user experience.

The system architecture ensures scalability, performance, and security, making it suitable for real-world applications such as corporate communication, educational collaboration, and productivity-focused environments. This project demonstrates the effective integration of **real-time systems, web technologies, and artificial intelligence** to develop a next-generation communication platform.

Overall, CharchaDesk highlights the potential of combining intelligent features with communication systems to create a more efficient, user-centric, and productivity-driven digital interaction environment.

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CHAPTER 1: INTRODUCTION

1.1 Overview

In the modern digital era, real-time communication has become an integral part of daily life, playing a crucial role in both personal and professional interactions. With the rapid advancement of internet technologies, chat applications have evolved significantly, enabling instant messaging, media sharing, and global connectivity. However, most existing communication platforms primarily focus on message exchange and lack intelligent features that enhance user productivity and manage distractions effectively.

The increasing dependency on digital communication has also introduced challenges such as information overload, unnecessary notifications, lack of focus, and limited administrative control. Users often struggle to prioritize important conversations, which affects their efficiency and workflow. This creates a need for a smarter communication system that not only enables real-time interaction but also enhances user productivity through intelligent features.

The proposed system, CharchaDesk Real-Time Chat Application, is designed to address these challenges by integrating real-time communication with advanced productivity and AI-driven functionalities. The system allows users to communicate through one-to-one messaging using WebSockets, ensuring low-latency and seamless interaction. It is developed using modern technologies such as Django for backend processing, PostgreSQL for database management, and Tailwind CSS for frontend design, providing a scalable and user-friendly platform.

CharchaDesk incorporates several essential and advanced features, including instant messaging, file sharing, message reactions, chat history management, and user safety mechanisms such as **blocking and reporting**. A key innovation of the system is the introduction of **Focus Mode**, which enables users to prioritize specific conversations while muting less important notifications, thereby reducing distractions and improving productivity.

Furthermore, the system integrates **Artificial Intelligence (AI) techniques**, particularly **Natural Language Processing (NLP)**, to enhance communication efficiency. Features such as **chat summarization and smart reply suggestions** help users quickly understand conversations and respond efficiently, saving time and effort. These intelligent capabilities make the system more adaptive and user-centric compared to traditional chat applications.

The architecture of CharchaDesk ensures **high performance, scalability, and security**, making it suitable for various real-world applications such as corporate communication, educational collaboration, and productivity-focused environments. The system demonstrates the effective combination of **real-time technologies, web development frameworks, and artificial intelligence** to create a next-generation communication platform. Overall, this project aims to bridge the gap between conventional messaging systems and intelligent productivity tools by delivering a secure, efficient, and feature-rich chat application that enhances both communication and user experience.

1.2 Objective and Scope

Objective:

The primary objective of this project is to develop an intelligent, scalable, and real-time chat application that enhances communication efficiency while integrating productivity-oriented and AI-driven features. The system aims to:

- **Enable Real-Time Communication:** Design and implement a system that supports instant messaging between users using WebSocket technology for low-latency interaction.
- **Enhance User Productivity:** Introduce features such as Focus Mode to help users prioritize important conversations and minimize distractions caused by unnecessary notifications.
- **Integrate Artificial Intelligence:** Incorporate AI-based functionalities such as chat summarization and smart reply suggestions using Natural Language Processing (NLP) techniques.
- **Ensure Security and User Safety:** Provide secure authentication, user data protection, and features like blocking and reporting to maintain a safe communication environment.
- **Develop a Scalable and Efficient System:** Build the application using modern technologies such as Django, PostgreSQL, and WebSockets to ensure high performance, scalability, and reliability.
- **Improve User Experience:** Create a responsive and user-friendly interface that simplifies communication and enhances overall usability.

Scope:

The scope of this project includes the complete design, development, and implementation of a web-based real-time chat application with intelligent features. The system covers:

- **User Authentication and Management:** Registration, login, and secure user handling.
- **Real-Time Messaging System:** Implementation of one-to-one chat using WebSocket communication.
- **Advanced Communication Features:** File sharing, message reactions, and chat history management.
- **AI-Based Enhancements:** Chat summarization and smart reply generation for improved efficiency.
- **Focus Mode Functionality:** Allowing users to control notifications and prioritize conversations.
- **Admin Control and Monitoring:** Tools for managing users, handling reports, and maintaining system integrity.
- **Scalability for Future Enhancements:** The system can be extended to include features such as group chats, voice/video calling, mobile applications, and advanced AI-based chatbots.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Existing Systems

The current communication systems include widely used chat applications such as WhatsApp, Telegram, Messenger, and professional platforms like Slack and Microsoft Teams. These systems provide features like real-time messaging, media sharing, and group communication. While they are efficient for communication, most of them primarily focus on message exchange rather than enhancing user productivity or providing intelligent assistance.

2.2 Limitations of Existing Systems

Despite their popularity, existing chat systems have several limitations:

- Lack of productivity-focused features
- Excessive notifications causing distraction
- Limited use of AI for smart communication
- Privacy and security concerns in some platforms
- Limited admin control and customization

These limitations highlight the need for an intelligent chat system like CharchaDesk that improves both communication and productivity.

CHAPTER 3: REQUIREMENTS

3.1 Functional Requirements

The functional requirements define the core functionalities of the CharchaDesk system:

- **User Authentication:** The system should allow users to register, login, and manage their accounts securely.
 - **Real-Time Messaging:** The system must support instant messaging between users using WebSocket communication.
 - **Message Management:** Users should be able to send, receive, and view chat history efficiently.
 - **File Sharing:** The system should allow users to share files and media within chats.
 - **AI-Based Features:** The system should provide smart reply suggestions and chat summarization using AI techniques.
 - **Focus Mode:** Users should be able to prioritize important conversations and mute unnecessary notifications.
 - **Admin Control:** The system should allow monitoring of users, handling reports, and managing misuse.
-

3.2 Non-Functional Requirements

These define the performance and quality attributes of the system:

- **Performance:** The system should provide fast and real-time message delivery with minimal delay.
- **Reliability:** The application should function consistently without failures.
- **Scalability:** The system should support future enhancements such as group chats and mobile applications.
- **Usability:** The user interface should be simple, responsive, and user-friendly.
- **Security:** User data and communication should be protected using secure authentication and protocols.
- **Availability:** The system should be accessible whenever required with minimal downtime.

3.3 Software and Hardware Requirements

Software Requirements

- **Operating System:** Windows 10/11 or Linux
 - **Frontend:** HTML, CSS, JavaScript, Tailwind CSS
 - **Backend:** Python, Django, Django REST Framework
 - **Programming Languages:**
 - Python (Backend & AI Integration)
 - JavaScript (Frontend)
 - **Database:** PostgreSQL
 - **Real-Time Communication:** WebSockets, Django Channels
 - **Libraries/Technologies:**
 - NLP (Natural Language Processing)
 - HuggingFace Transformers (for AI features)
 - **Development Tools:**
 - VS Code / PyCharm
 - Git (Version Control)
 - Postman (API Testing)
 - Web Browser (Chrome/Edge)
-

Hardware Requirements

- **Computer/Laptop:** Minimum Intel i3 processor
- **RAM:** Minimum 4GB (8GB recommended)
- **Storage:** At least 20GB free space
- **Internet Connection:** Stable internet for real-time communication
- **Peripherals:** Keyboard, Mouse, Monitor

CHAPTER 4: SYSTEM DESIGN And IMPLEMENTATION

4.1 System Architecture

The proposed system architecture of **CharchaDesk Real-Time Chat Application** integrates frontend, backend, database, and real-time communication components to provide seamless and efficient messaging.

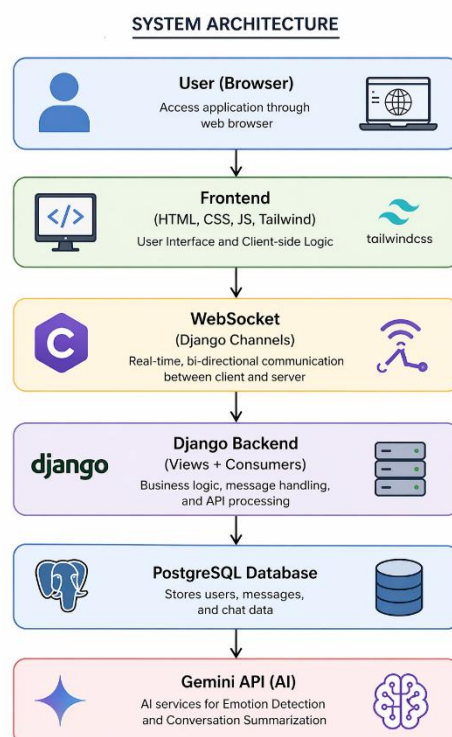
The system begins with the **user interface**, developed using modern web technologies, where users interact with the application by sending and receiving messages. The frontend communicates with the backend server through HTTP requests and WebSocket connections.

The **backend**, developed using Django and Django REST Framework, handles user authentication, message processing, and system logic. For real-time communication, **WebSockets (Django Channels)** are used to ensure instant message delivery with minimal latency.

All user data, chat messages, and related information are stored in the **PostgreSQL database**, ensuring efficient data management and retrieval.

Additionally, the system integrates an **AI module** that processes chat data using Natural Language Processing (NLP) techniques to provide features such as chat summarization and smart reply suggestions.

The processed data is then sent back to the frontend, where it is displayed to the user in an interactive and user-friendly format.



System Flow:

User → Frontend Interface → WebSocket/HTTP Request → Django Backend → Database / AI Module → Response → Frontend Display

4.2 System Environment / Deployment Design

The CharchaDesk system is a web-based application and does not require specialized hardware components. It operates on a standard client-server architecture using commonly available computing devices.

System Components:

Component	Description
Client Device	User device (Laptop/Desktop) to access application
Web Browser	Interface for accessing the chat system
Application Server	Django backend handling logic and APIs
Database Server	PostgreSQL database for data storage
Network	Internet connection for real-time communication

System Configuration:

Component Requirement

Processor	Intel i3 or higher
RAM	Minimum 4GB (8GB recommended)
Storage	20GB free space
Browser	Chrome / Edge / Firefox
Internet	Stable broadband connection

The system enables users to access the application through a web browser, where requests are processed by the Django backend. Real-time communication is handled using WebSockets, ensuring instant message delivery. All data, including user information and chat history, is securely stored in the PostgreSQL database.

4.3 Software Design

The software system of CharchaDesk consists of three main components:

1. Frontend System

The frontend is developed using HTML, CSS, JavaScript, and Tailwind CSS. It provides an interactive user interface for communication.

Features:

- Real-time chat interface
- Message display and input
- Notifications and Focus Mode
- User-friendly design

2. Backend System (Django)

The backend is developed using Django and Django REST Framework. It handles application logic and communication.

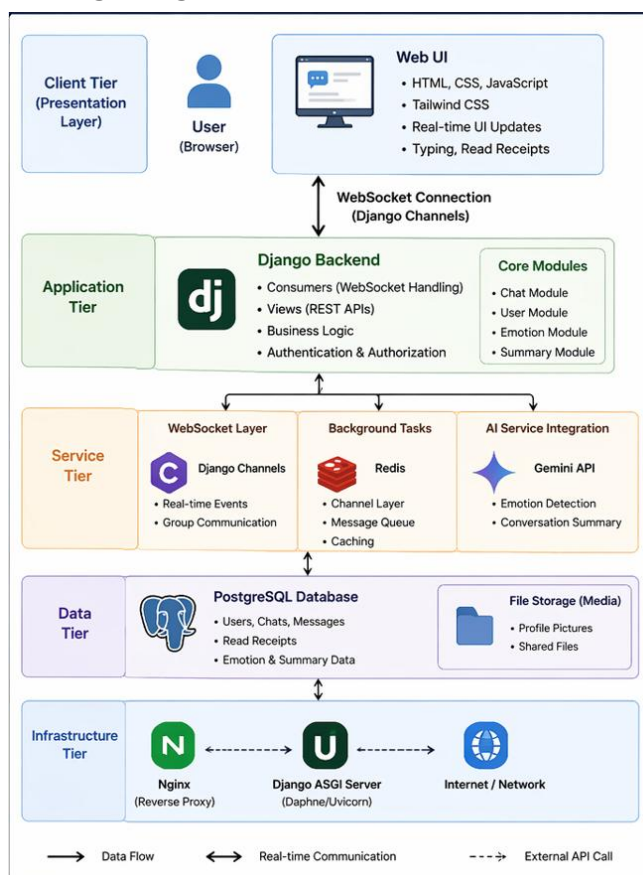
Responsibilities:

- User authentication and authorization
- Message processing and storage
- API handling
- Integration with WebSockets for real-time communication

3. Database System (PostgreSQL)

The database stores user data, messages, and chat history efficiently for quick retrieval and scalability.

High Level Design Diagram -



4.4 Implementation Details

The implementation involves integration of frontend, backend, database, and real-time communication.

Step 1: User Authentication

Users register and log in securely using the backend system.

Step 2: Real-Time Communication

Messages are sent and received using WebSocket connections for instant delivery.

Step 3: Message Processing

Backend processes messages and stores them in the database.

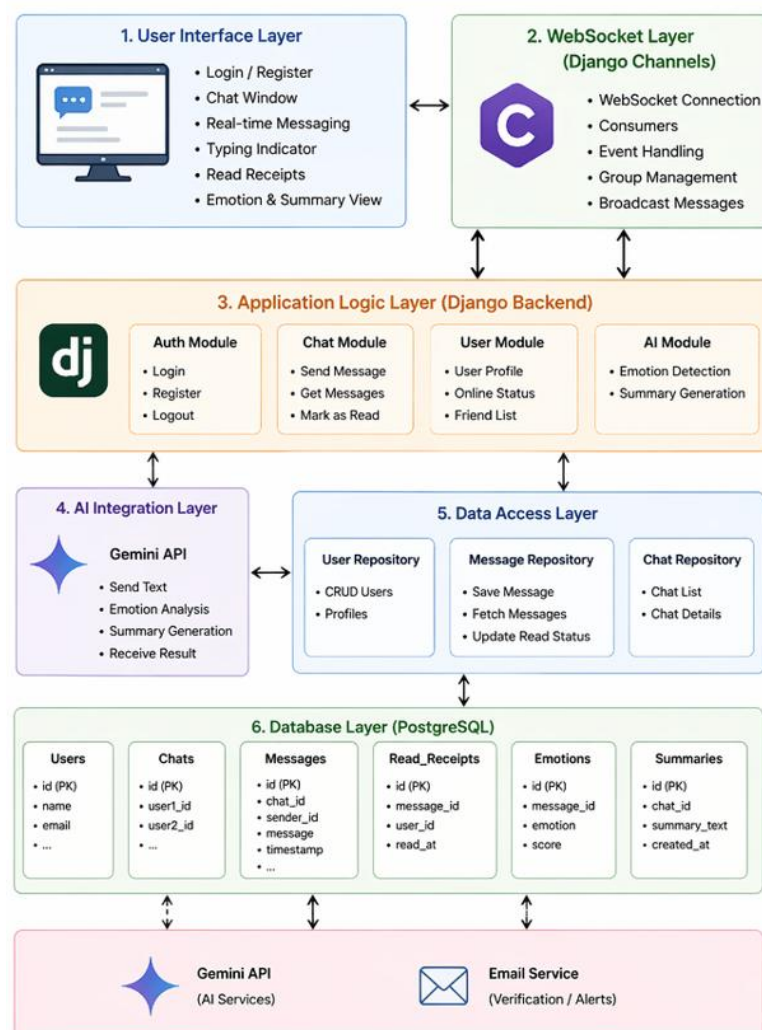
Step 4: AI Processing

Chat data is analyzed using NLP techniques for summarization and smart replies.

Step 5: Display Output

Processed messages and AI suggestions are displayed on the user interface.

Low Level Design diagram -



4.5 Data Handling

The system manages data such as user information, chat messages, and interaction logs.

Data includes:

- User details (name, email, credentials)
- Messages and timestamps
- Chat history

Data is stored in PostgreSQL and can be used for analytics and AI processing.

4.6 AI Module

The AI module enhances communication using Natural Language Processing (NLP).

Input:

- Chat messages

Output:

- Chat summaries
- Smart reply suggestions

Technologies used:

- NLP techniques
 - Transformers (HuggingFace)
-

4.7 API Design

The backend provides REST APIs for communication between frontend and server.

Example APIs:

POST /send-message

Input:

```
{  
  "message": "Hello"  
}
```

GET /chat-history

Output:

```
{  
  "messages": [...]  
}
```


4.8 User Interface Design

The user interface is designed to be simple, responsive, and interactive.

Features:

- Chat window for messaging
- Notification system
- Focus Mode control
- Display of chat history and AI suggestions

The interface ensures ease of use and enhances overall user experience.

CHAPTER 5: TESTING AND RESULTS

5.1 Testing Methodology

To ensure the proper functioning of the CharchaDesk system, different levels of testing were performed on the frontend, backend, and real-time communication components.

The testing process verifies whether the system correctly handles user authentication, message transmission, data storage, and AI-based features.

The following testing approaches were used:

- **Frontend Testing:** The user interface was tested to ensure proper display of messages, responsiveness, and smooth user interaction.
- **Backend Testing:** Django backend APIs were tested using Postman to verify correct handling of requests and responses.
- **WebSocket Testing:** Real-time messaging functionality was tested to ensure instant message delivery without delay.
- **AI Module Testing:** AI features such as chat summarization and smart reply suggestions were tested for accuracy and relevance.
- **Integration Testing:** End-to-end testing was performed to ensure proper communication between frontend, backend, database, and AI module.

5.2 Test Cases and Results

Test Case ID	Module	Test Description	Expected Result	Status
TC-01	Authentication	User login and registration	User authenticated successfully	Pass
TC-02	Messaging	Send and receive messages	Messages delivered instantly	Pass
TC-03	WebSocket	Real-time communication	No delay in message transmission	Pass
TC-04	API	Backend API response	Valid response returned	Pass
TC-05	AI Module	Smart reply and summarization	Relevant output generated	Pass
TC-06	Integration	Complete system workflow	System works correctly	Pass

5.3 Result Analysis

The results obtained from the system show that CharchaDesk successfully enables real-time communication with integrated intelligent features.

Observations:

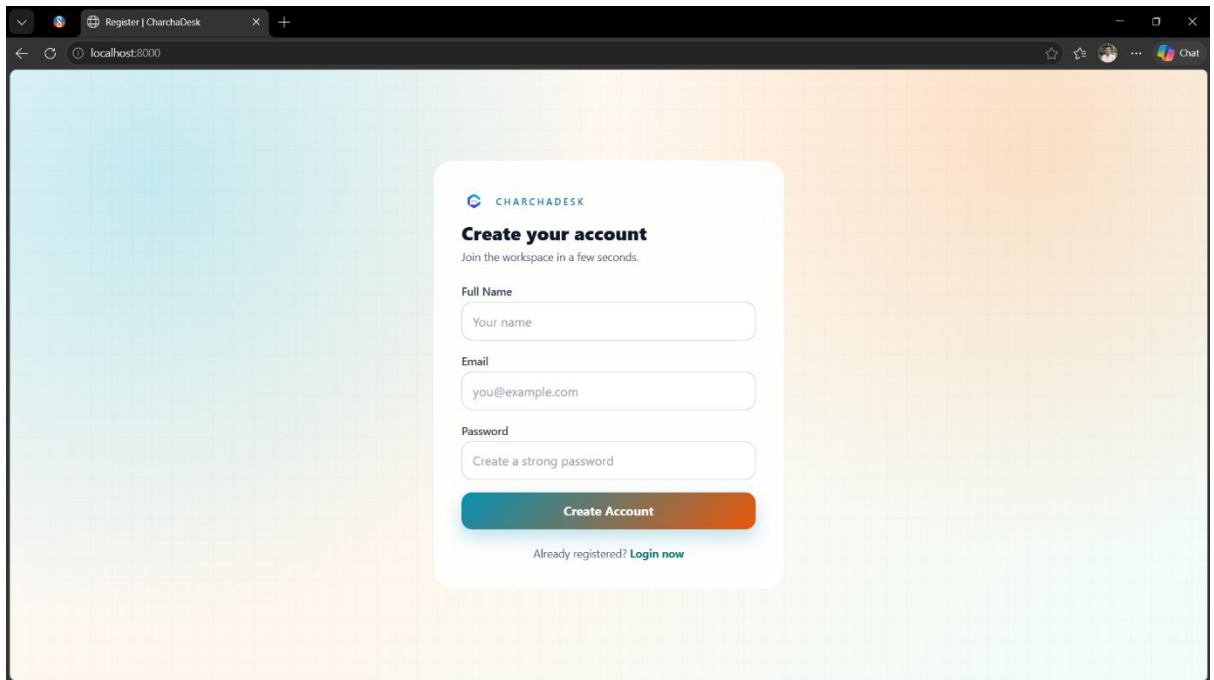
- Messages are delivered instantly using WebSocket technology.
 - Backend APIs handle requests efficiently.
 - AI features provide useful suggestions and summaries.
 - User interface displays data correctly and smoothly.
 - The system performs reliably under normal usage conditions.
-

5.4 Advantages of System

- Real-time communication with low latency
- Improved productivity through Focus Mode
- AI-based smart reply and summarization
- Secure and scalable architecture
- User-friendly interface
- Suitable for future enhancements

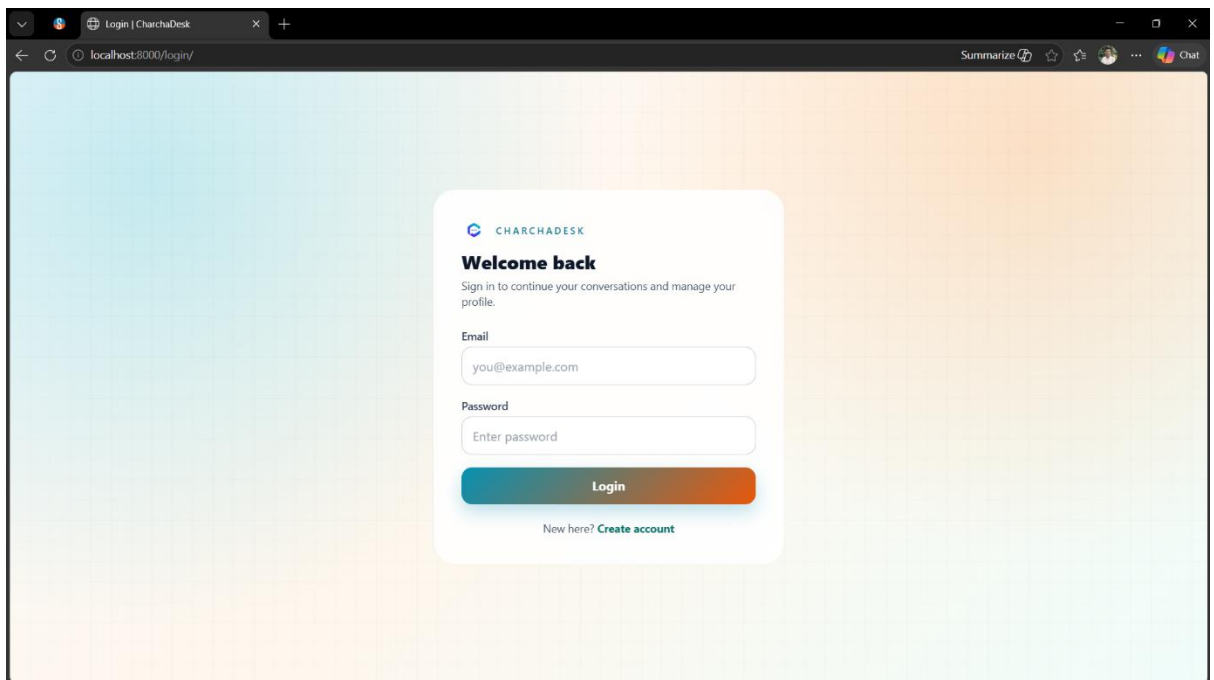
CHAPTER 6: APPENCES

Registratio page-



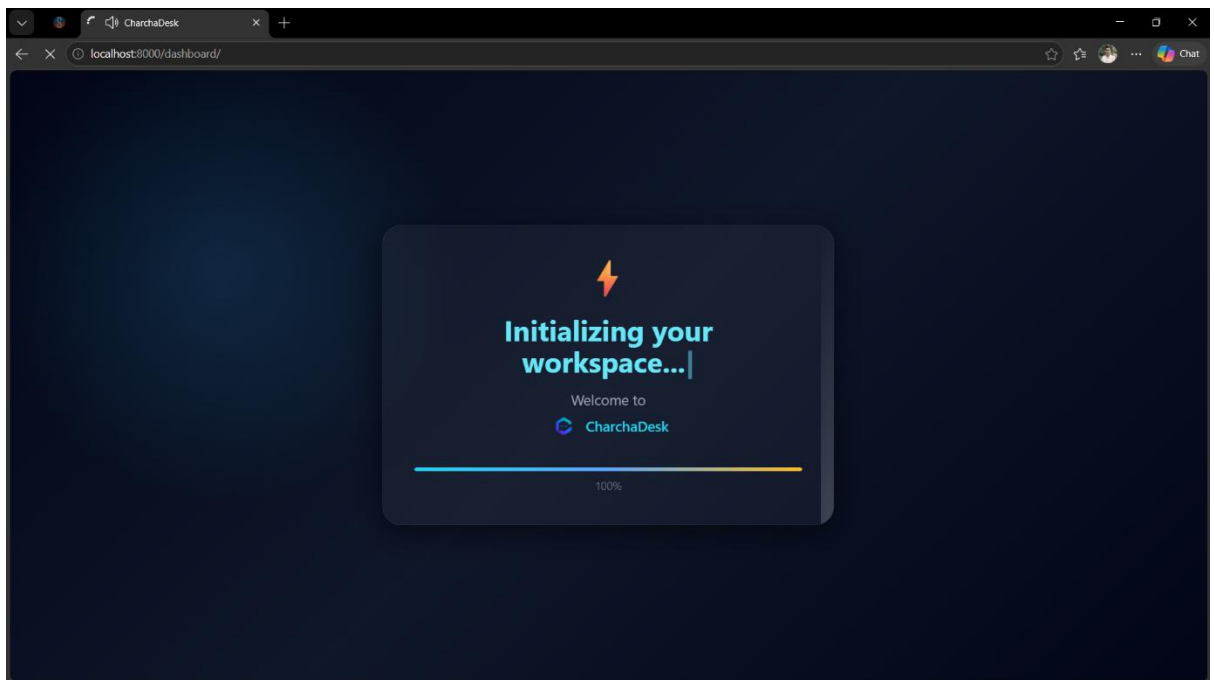
The screenshot shows a web browser window with the title "Register | CharchaDesk". The address bar shows "localhost:8000". The page features a registration form titled "Create your account" with the subtitle "Join the workspace in a few seconds." The form includes three input fields: "Full Name" (placeholder: "Your name"), "Email" (placeholder: "you@example.com"), and "Password" (placeholder: "Create a strong password"). Below the fields is a blue "Create Account" button. At the bottom, there is a link: "Already registered? [Login now](#)".

Login Page –

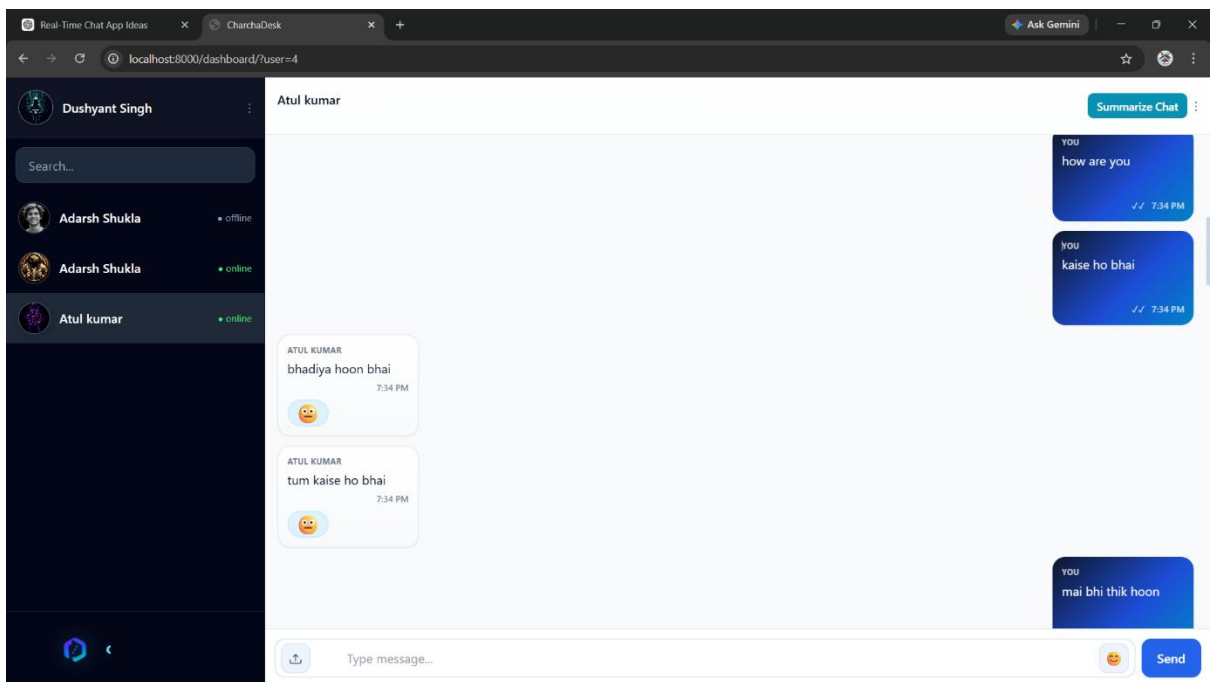


The screenshot shows a web browser window with the title "Login | CharchaDesk". The address bar shows "localhost:8000/login/". The page features a login form titled "Welcome back" with the subtitle "Sign in to continue your conversations and manage your profile." The form includes two input fields: "Email" (placeholder: "you@example.com") and "Password" (placeholder: "Enter password"). Below the fields is a blue "Login" button. At the bottom, there is a link: "New here? [Create account](#)".

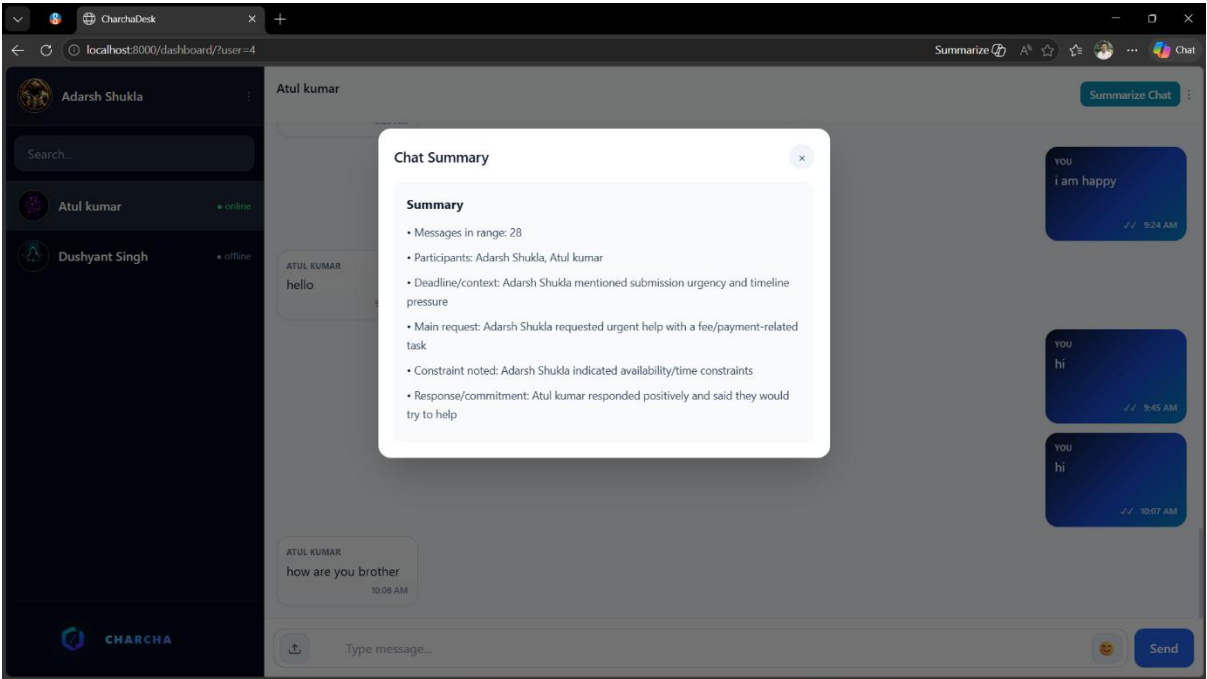
Welcome Page-



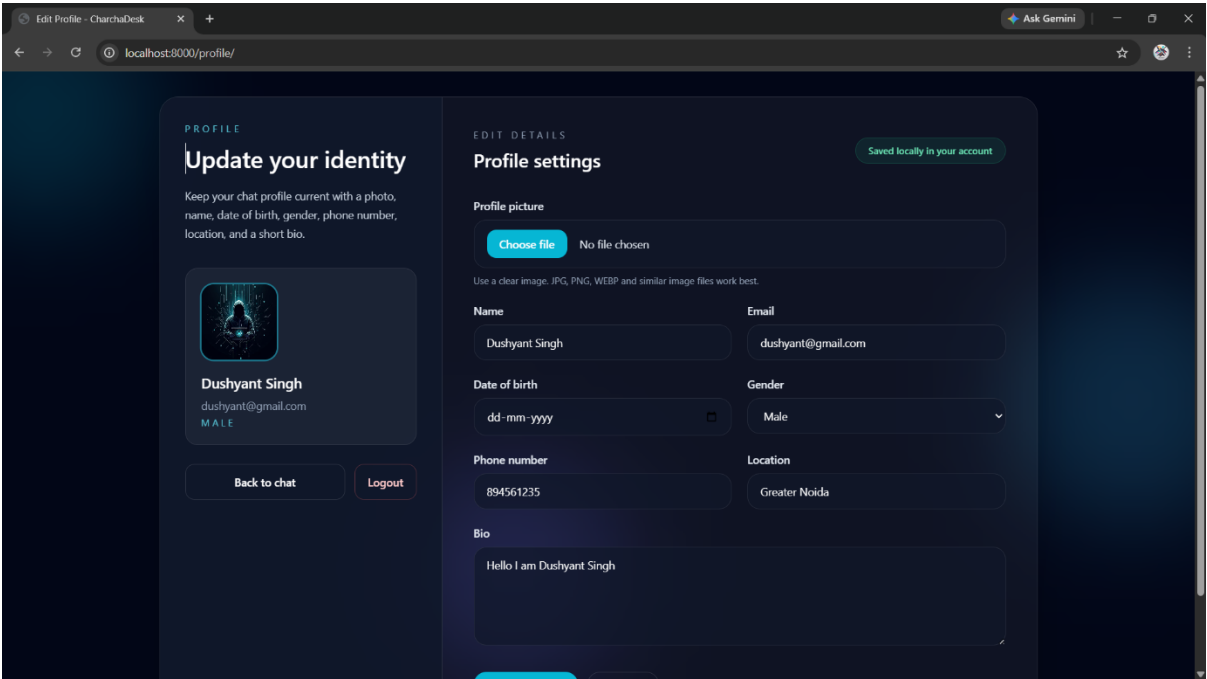
Dashboard page –



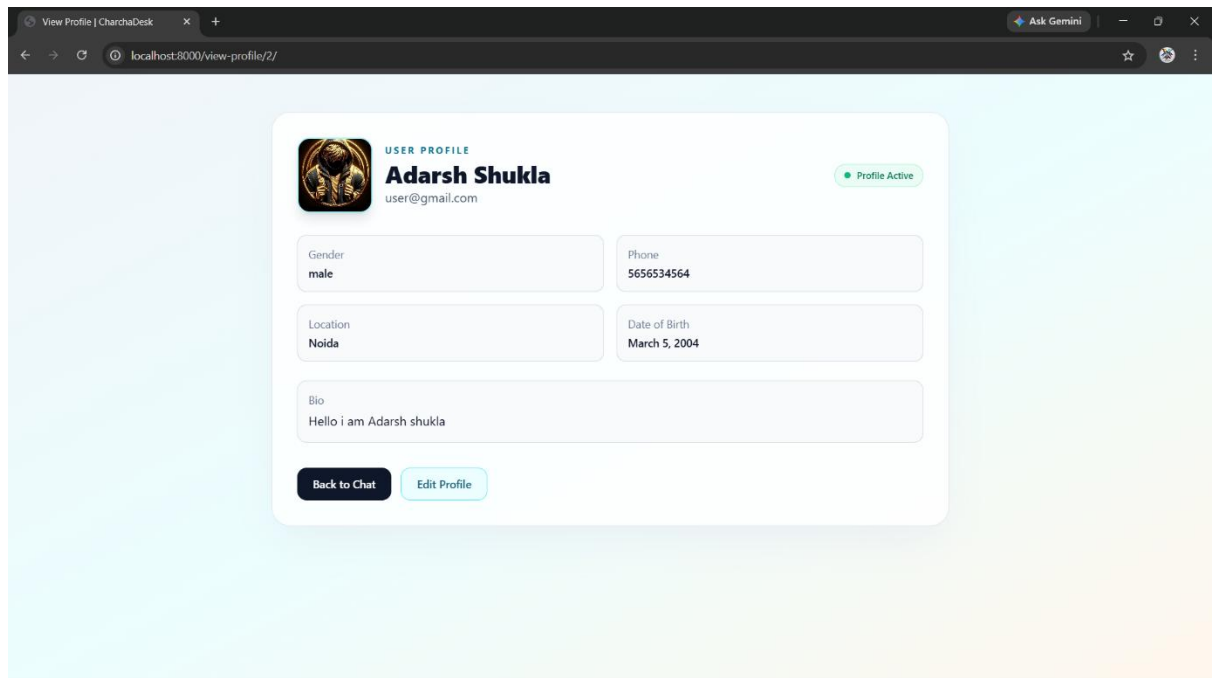
Chat Summary Page –



Profile Edit Page –



Profile View Page –



CHAPTER 7:

CONCLUSION AND FUTURE WORK

7.1: CONCLUSION

The proposed system, **CharchaDesk Real-Time Chat Application**, successfully demonstrates the development of a modern communication platform that integrates real-time messaging with intelligent and productivity-enhancing features. Unlike traditional chat applications that focus only on message exchange, this system provides a more efficient and user-centric approach to communication.

The application is developed using technologies such as **Django, PostgreSQL, and WebSockets**, ensuring fast, reliable, and scalable performance. Features like **real-time messaging, file sharing, chat history, and user safety mechanisms** provide a complete communication solution. Additionally, the integration of **AI-based functionalities**, including chat summarization and smart reply suggestions, enhances user experience and efficiency.

The implementation of **Focus Mode** helps users manage distractions and prioritize important conversations, making the system suitable for both personal and professional use. The results indicate that the system performs efficiently with low latency and reliable communication.

Overall, the project highlights the effective integration of **web technologies, real-time systems, and artificial intelligence** to create a next-generation chat application that improves both communication and productivity.

7.2: Future Work

Although the proposed system achieves its objectives, there are several opportunities for further enhancement:

- **Mobile Application Development:** Extend the system to Android and iOS platforms for better accessibility.
- **Voice and Video Communication:** Integrate real-time audio and video calling features.
- **Advanced AI Features:** Implement more advanced AI capabilities such as chatbot assistance and sentiment analysis.
- **Cloud Integration:** Use cloud platforms for scalable deployment and data storage.
- **End-to-End Encryption:** Enhance security by implementing stronger encryption techniques.
- **Group Chat and Collaboration Tools:** Add support for group communication and collaborative features.
- **Advanced Analytics Dashboard:** Provide insights and analytics for user activity and communication trends.

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APPENDICES

Source Code

The source code of the proposed system, CharchaDesk – Real-Time Chat Application, is organized in a modular structure consisting of backend services, real-time communication modules, database models, and frontend user interface components. This modular design ensures maintainability, scalability, and efficient development.

The system integrates Django backend, WebSocket-based real-time communication, PostgreSQL database, and AI-based features such as chat summarization and smart reply suggestions. Each component is designed to handle a specific functionality within the system, ensuring smooth communication and performance.

Key File Highlights

- frontend/index.html

Contains the main structure of the chat application interface including message display and input fields.

- frontend/styles.css / tailwind.config.js

Defines the styling of the application using CSS and Tailwind CSS for responsive and modern UI design.

- frontend/app.js

Handles client-side logic such as sending messages, receiving responses, and updating the UI dynamically.

- backend/views.py

Handles HTTP requests and responses, processes chat messages, and connects frontend with backend logic.

- backend/urls.py

Defines API routes such as message sending and chat history retrieval.

- backend/models.py

Defines database models for users, messages, and chat history.

- backend/consumers.py

Implements WebSocket consumers for real-time communication between users.

- backend/serializers.py

Serializes and deserializes data for REST API communication using Django REST Framework.

- backend/settings.py

Contains project configuration including database settings, installed apps, and middleware.

- database/schema.sql (PostgreSQL)

Defines the database schema for storing user data, messages, and logs.

- ai_module/nlp_processor.py

Processes chat messages using NLP techniques for summarization and smart reply generation.

- ai_module/model_integration.py

Integrates transformer-based models (HuggingFace) with backend services.

- frontend/src/services/api.js

Handles communication between frontend **and backend using REST APIs.**